

HUMBER ENGINEERING

MENG-3020

SYSTEMS MODELING & SIMULATION

LECTURE 3

LECTURE 3

State-Space Approach to Modeling Dynamic Systems

- Standard Forms for System Models
- State Variable Equations
- Vector-Matrix Form of State-Variable Models
 - Standard Form of The State Equation
 - The Output Equation
- Block Diagram Models

Standard Forms for System Models

- In the previous lecture, we introduced the element and interconnection laws for **translational mechanical** systems and the procedure for drawing free-body diagrams and applying Newton's second law to find the **equation of motion**.
- These are **essential** steps regardless of the final form of the equation.
- However, the mathematical model should have a **standard form**
- **Why do we need consider standard forms?**
 - More convenient to find the solution of the equation
 - Easy to compare the model to another
 - Necessary for entry into a computer solver, such as MATLAB / Simulink
 - Different standard forms have different usages and purposes
- The two most common standard forms for dynamic system models:
 - **State-Space Model**
 - Provides a model in **time domain**, and **internal description** of the system via state variables
 - Applicable for time-varying, nonlinear and MIMO systems
 - **Transfer Function Model**
 - Provides a model in **Laplace domain**, and input-output description (**External description**)
 - Limited to LTI and SISO or two-input two-output systems

State-Space Representation of Dynamic Systems

- In **State Space Representation**, dynamic model of the system is described by a set of first-order differential equations in terms of the variables called the **state variables**.
- Formulating the model in state-space form is to begin by **selecting** a set of **state variables**.

□ Definition of State Variables

- The **state variables** of a dynamic system are the **minimum set of linearly independent variables** that describes the effect of the history of the system (past inputs and dynamics) on its response in the future.
- The **minimum number** of required state variables equals the order of the differential equation describing the system.
- Note that there is **no unique** set of state variables that describe any given system; many different sets of state variables may be selected to obtain a complete system description.
- The **state variables** may be selected based on **physical** and **measurable variables**, or in terms of variables that **are not directly measurable**.
- In physical dynamic systems it is often convenient to associate the **state variables** with the **energy storage elements** in the system. Because any energy that is initially stored in these elements can affect the response of the system at a later time.

State Variable Selection

- Table of **energy storage elements** in physical dynamic systems:

System	Element	Energy	Physical Variable
Electrical Systems	Capacitor C	$\frac{1}{2} C v^2$	Voltage $v(t)$
	Inductor L	$\frac{1}{2} L i^2$	Current $i(t)$
Translational Mechanical Systems	Mass M	$\frac{1}{2} M v^2$	Translational Velocity $v(t)$
	Translational Spring K	$\frac{1}{2} K x^2$	Translational Displacement $x(t)$
Rotational Mechanical Systems	Moment of Inertia J	$\frac{1}{2} J \omega^2$	Angular Velocity $\omega(t)$
	Tortional Spring K	$\frac{1}{2} K \theta^2$	Angular Displacement $\theta(t)$

State Variable Equations

Example 1

Find the state-variable equations for the given mass-spring-damper system.
Assume the applied force $f(t)$ as the input, and the displacement $x(t)$ is the output.

Draw the free-body diagram of the system and write the equation of motion of the system.

$$f(t) - Kx(t) - B\dot{x}(t) = M\ddot{x}(t)$$

The **state variables** q_1 and q_2 are selected as the **displacement of the spring $x(t)$** and **velocity of the mass $\dot{x}(t)$** .

$$q_1(t) = x(t) \rightarrow \dot{q}_1(t) = \dot{x}(t) \rightarrow \dot{q}_1(t) = q_2(t) \quad \text{Eqn. (1)}$$

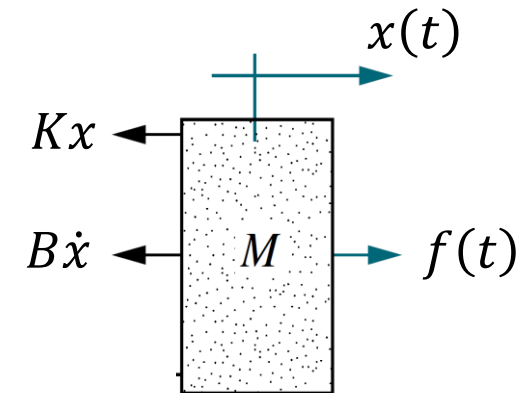
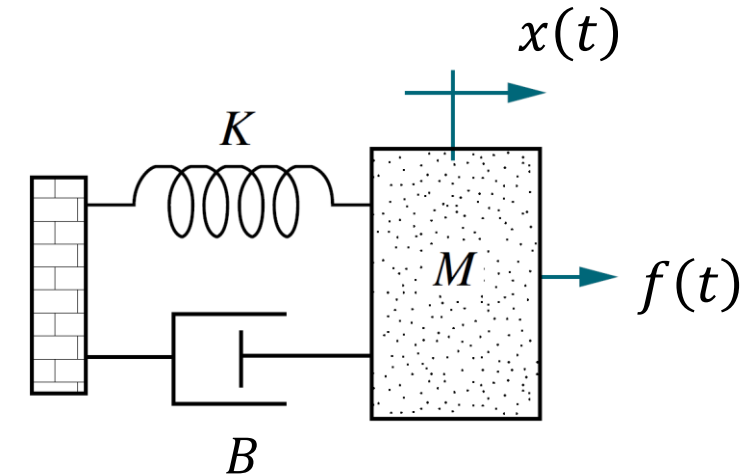
$$\begin{aligned} q_2(t) = \dot{x}(t) \rightarrow \dot{q}_2(t) = \ddot{x}(t) \rightarrow \dot{q}_2(t) &= \frac{1}{M}f(t) - \frac{K}{M}x(t) - \frac{B}{M}\dot{x}(t) \\ &\rightarrow \dot{q}_2(t) = \frac{1}{M}f(t) - \frac{K}{M}q_1(t) - \frac{B}{M}q_2(t) \quad \text{Eqn. (2)} \end{aligned}$$

State-variable equations are obtained as:

$$\begin{cases} \dot{q}_1(t) = q_2(t) \\ \dot{q}_2(t) = -\frac{K}{M}q_1(t) - \frac{B}{M}q_2(t) + \frac{1}{M}f(t) \end{cases}$$

The **output equation** is obtained as:

$$y(t) = q_1(t)$$

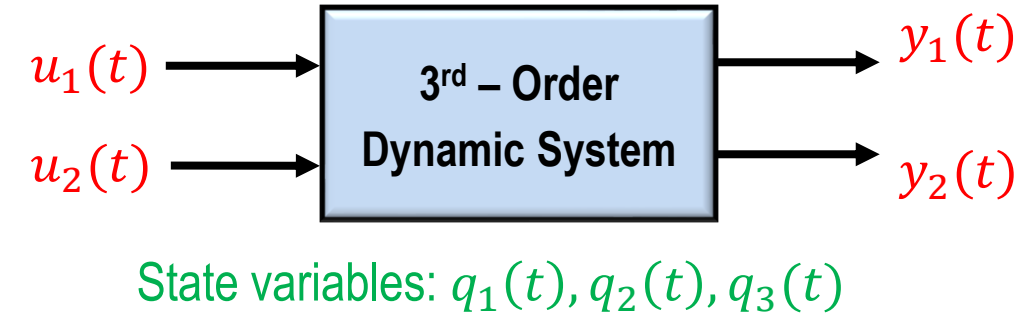


Free-body Diagram

General Form of The State-Space Equations

- Consider a 3rd order LTI dynamic system with 2 inputs, 2 outputs and 3 state-variables.

- Inputs** are the external influences on the system (force, voltage, ...)
- Outputs** are the variables of interest to be measured or calculated (position, velocity, current, ...)
- State variables** represent the status or memory of the system



- General form of a **State Space Model** for this system includes **state equations** and **output equations**:

State Equations

$$\begin{cases} \dot{q}_1 = a_{11}q_1 + a_{12}q_2 + a_{13}q_3 + b_{11}u_1 + b_{12}u_2 \\ \dot{q}_2 = a_{21}q_1 + a_{22}q_2 + a_{23}q_3 + b_{21}u_1 + b_{22}u_2 \\ \dot{q}_3 = a_{31}q_1 + a_{32}q_2 + a_{33}q_3 + b_{31}u_1 + b_{32}u_2 \end{cases}$$



$$\begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\dot{\mathbf{q}}(t) = \mathbf{A}\mathbf{q}(t) + \mathbf{B}\mathbf{u}(t)$$

Output Equations

$$\begin{cases} y_1 = c_{11}q_1 + c_{12}q_2 + c_{13}q_3 + d_{11}u_1 + d_{12}u_2 \\ y_2 = c_{21}q_1 + c_{22}q_2 + c_{23}q_3 + d_{21}u_1 + d_{22}u_2 \end{cases}$$



$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} + \begin{bmatrix} d_{11} & d_{12} \\ d_{21} & d_{22} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{q}(t) + \mathbf{D}\mathbf{u}(t)$$

Vector-Matrix Form of The State-Space Equations

- For **LTI systems**, the state and output equations can be written in **vector-matrix** form.
- General form of a State Space Representation for a n th order LTI system with m input and p output is:

$$\begin{cases} \dot{\mathbf{q}}(t) = \mathbf{A}\mathbf{q}(t) + \mathbf{B}\mathbf{u}(t) & \leftarrow \text{state equation} \\ \mathbf{y}(t) = \mathbf{C}\mathbf{q}(t) + \mathbf{D}\mathbf{u}(t) & \leftarrow \text{output equation} \end{cases}$$

$\mathbf{A}_{n \times n}$: system matrix

$\mathbf{B}_{n \times m}$: input matrix

$\mathbf{q}(t)$: state vector

$\mathbf{C}_{p \times n}$: output matrix

$\mathbf{D}_{p \times m}$: feed-forward matrix

$\mathbf{u}(t)$: input vector

$\mathbf{y}(t)$: output vector

$$\begin{matrix} \swarrow \\ \mathbf{q}(t) = \begin{bmatrix} q_1(t) \\ q_2(t) \\ \vdots \\ q_n(t) \end{bmatrix} \\ \swarrow \\ n \times 1 \text{ vector} \end{matrix}, \quad \begin{matrix} \swarrow \\ \mathbf{u}(t) = \begin{bmatrix} u_1(t) \\ u_2(t) \\ \vdots \\ u_m(t) \end{bmatrix} \\ \swarrow \\ m \times 1 \text{ vector} \end{matrix}, \quad \begin{matrix} \swarrow \\ \mathbf{y}(t) = \begin{bmatrix} y_1(t) \\ y_2(t) \\ \vdots \\ y_p(t) \end{bmatrix} \\ \swarrow \\ p \times 1 \text{ vector} \end{matrix}$$

- $\dot{\mathbf{q}}(t)$ is the **derivative of the state vector**, which shows how the state vector changes as a linear combination of the state vector and the input vector.
- In **LTI** systems $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$ and $\mathbf{D}$ are **constant** matrices.
- In **time-varying linear systems** some elements of the matrices would be **functions of time**.

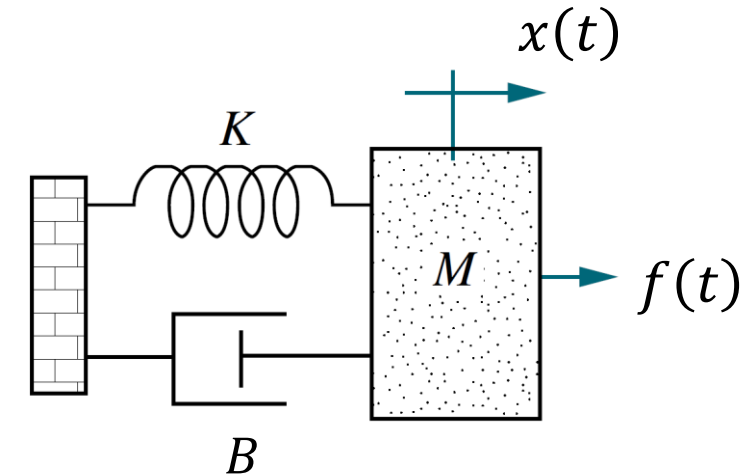
Vector-Matrix Form of The State-Space Equations

Example 2

Find the vector-matrix form of the state-space equations for the given mass-spring-damper system. Assume the applied force $f(t)$ as the input, and the displacement $x(t)$ is the output.

Recall the state equations and the output equation from Example 1:

$$\begin{cases} \dot{q}_1(t) = q_2(t) \\ \dot{q}_2(t) = \frac{1}{M}f(t) - \frac{K}{M}q_1(t) - \frac{B}{M}q_2(t) \\ y(t) = q_1(t) \end{cases}$$



We can represent the **state** and **output equations** in the standard **matrix-vector form** as below:

$$\boxed{\dot{\mathbf{q}}(t) = \mathbf{A}\mathbf{q}(t) + \mathbf{B}u(t)} \quad \rightarrow \quad \begin{bmatrix} \dot{q}_1(t) \\ \dot{q}_2(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{K}{M} & -\frac{B}{M} \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{M} \end{bmatrix} f(t)$$

State Equation

$$\boxed{\mathbf{y}(t) = \mathbf{C}\mathbf{q}(t) + \mathbf{D}u(t)} \quad \rightarrow \quad y(t) = [1 \quad 0] \begin{bmatrix} q_1(t) \\ q_2(t) \end{bmatrix} + [0]f(t)$$

Output Equation

State-Space Representation

Example 3

Find the state-space model for the given mass-spring-damper system.

Assume the applied force $f_a(t)$ as the input, and the displacement of spring K_2 is the output.

Draw the free-body diagram of the system and write the equation of motion of the system.

$$\text{Mass } M_1 \rightarrow -K_1 x_1 - K_2(x_1 - x_2) - B(\dot{x}_1 - \dot{x}_2) = M_1 \ddot{x}_1$$

$$\text{Mass } M_2 \rightarrow f_a(t) - K_2(x_2 - x_1) - B(\dot{x}_2 - \dot{x}_1) = M_2 \ddot{x}_2$$

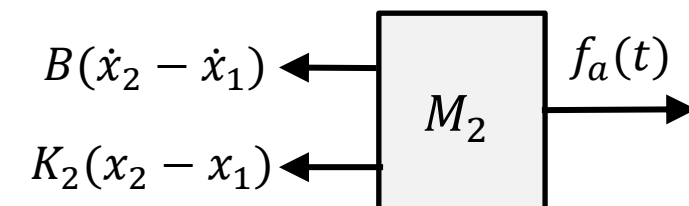
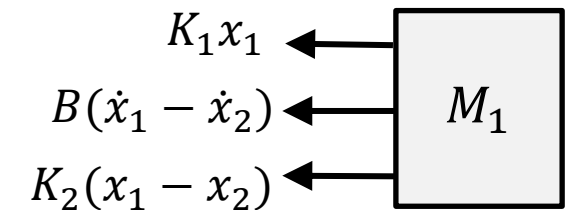
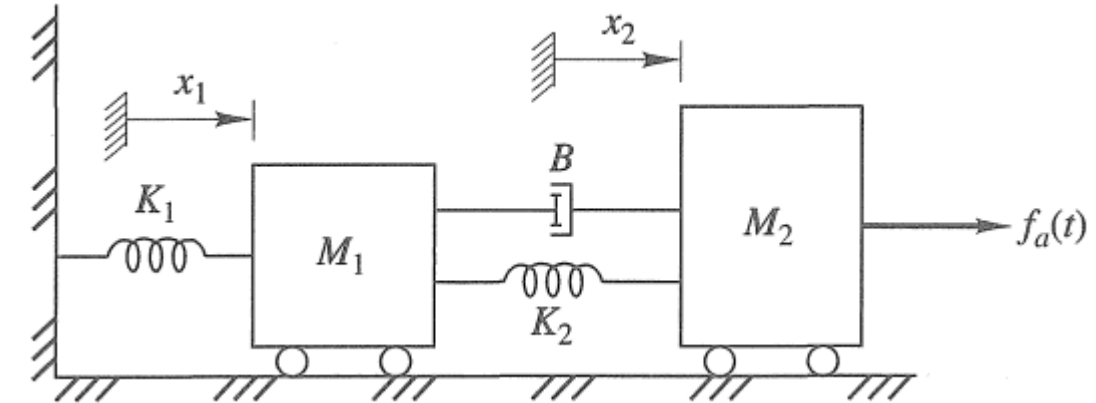
The **state variables** q_1, q_2, q_3 and q_4 are selected as the **displacement of the springs** and **velocity of the masses**.

$$q_1 = x_1 \rightarrow \dot{q}_1 = \dot{x}_1 \rightarrow \dot{q}_1 = q_2 \quad \text{Eqn. (1)}$$

$$\begin{aligned} q_2 = \dot{x}_1 &\rightarrow \dot{q}_2 = \ddot{x}_1 \rightarrow \dot{q}_2 = \frac{1}{M_1} (-K_1 x_1 - K_2(x_1 - x_2) - B(\dot{x}_1 - \dot{x}_2)) \\ &\rightarrow \dot{q}_2 = \frac{1}{M_1} (-K_1 q_1 + K_2 q_3 - B(q_2 - q_4)) \quad \text{Eqn. (2)} \end{aligned}$$

$$q_3 = x_2 - x_1 \rightarrow \dot{q}_3 = \dot{x}_2 - \dot{x}_1 \rightarrow \dot{q}_3 = q_4 - q_2 \quad \text{Eqn. (3)}$$

$$\begin{aligned} q_4 = \dot{x}_2 &\rightarrow \dot{q}_4 = \ddot{x}_2 \rightarrow \dot{q}_4 = \frac{1}{M_2} (f_a(t) - K_2(x_2 - x_1) - B(\dot{x}_2 - \dot{x}_1)) \\ &\rightarrow \dot{q}_4 = \frac{1}{M_2} (f_a(t) - K_2 q_3 - B(q_4 - q_2)) \quad \text{Eqn. (4)} \end{aligned}$$



State-Space Representation

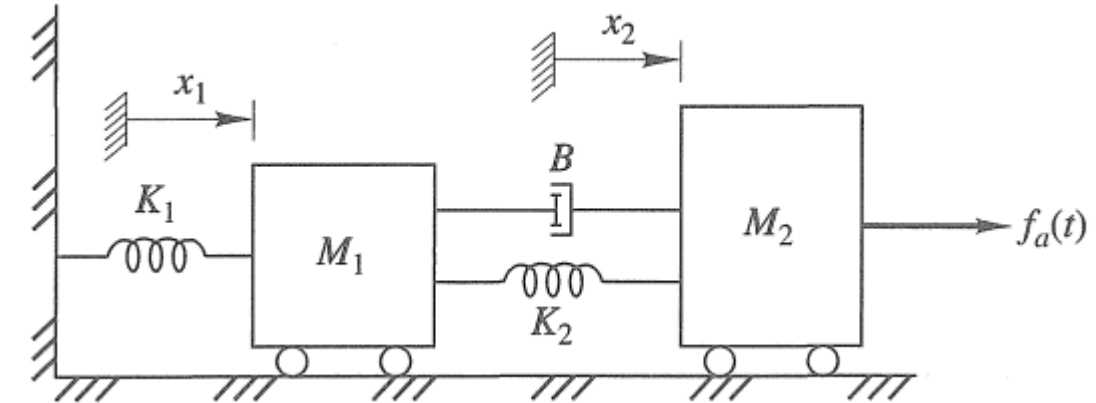
Example 3

Find the state-space model for the given mass-spring-damper system.

Assume the applied force $f_a(t)$ as the input, and the displacement of spring K_2 is the output.

The state-variable equations and output equation are obtained as:

$$\begin{cases} \dot{q}_1 = q_2 \\ \dot{q}_2 = \frac{1}{M_1}(-K_1 q_1 - B q_2 + K_2 q_3 + B q_4) \\ \dot{q}_3 = -q_2 + q_4 \\ \dot{q}_4 = \frac{1}{M_2}(B q_2 - K_2 q_3 - B q_4 + f_a(t)) \\ y = x_2 - x_1 \rightarrow y = q_3 \end{cases}$$



We can represent the state and output equations in the standard matrix-vector form as below:

State Equation $\dot{\mathbf{q}}(t) = \mathbf{A} \mathbf{q}(t) + \mathbf{B} \mathbf{u}(t)$

$\mathbf{y}(t) = \mathbf{C} \mathbf{q}(t) + \mathbf{D} \mathbf{u}(t)$ Output Equation

$$\begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \\ \dot{q}_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -K_1/M_1 & -B/M_1 & K_2/M_1 & B/M_1 \\ 0 & -1 & 0 & 1 \\ 0 & B/M_2 & -K_2/M_2 & -B/M_2 \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1/M_2 \end{bmatrix} f_a(t)$$

$$y(t) = [0 \quad 0 \quad 1 \quad 0] \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{bmatrix} + [0] u(t)$$

State-Space Representation

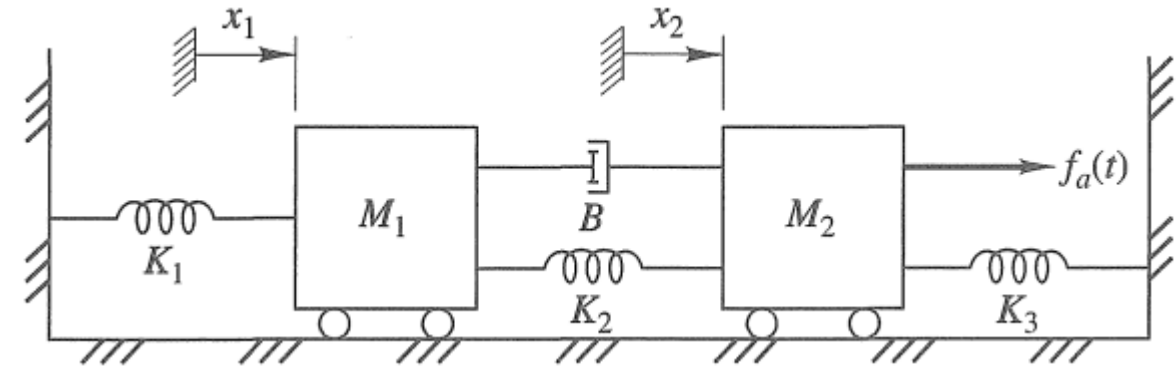
- The following example illustrates how the number of state variables can be **less** than the number of energy-storing elements.

Example 4 Find the state-space model for the following LTI system.
Assume the applied force $f_a(t)$ as the input, and the displacement of spring K_2 is the output.

Draw the free-body diagram of the system and write the equation of motion of the system.

Mass $M_1 \rightarrow -K_1x_1 - K_2(x_1 - x_2) - B(\dot{x}_1 - \dot{x}_2) = M_1\ddot{x}_1$

Mass $M_2 \rightarrow f_a(t) - K_2(x_2 - x_1) - B(\dot{x}_2 - \dot{x}_1) - K_3x_2 = M_2\ddot{x}_2$

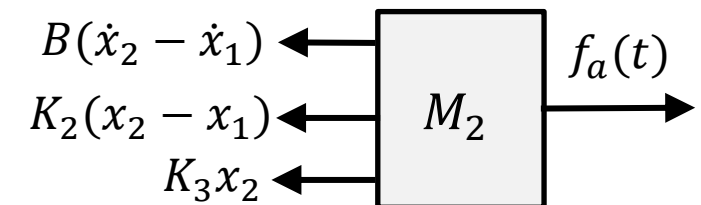
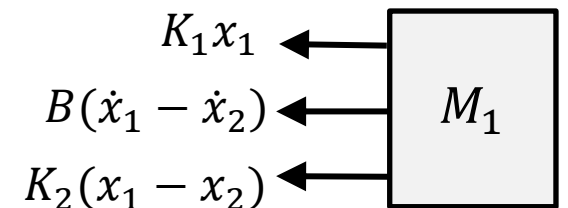


Although there are three springs, their elongations **are not all independent** and can be specified in terms of the two displacement variables x_1 and x_2 .

The elongations of K_1 , K_2 , and K_3 are x_1 , $x_2 - x_1$, and x_2 , respectively.

The **state variables** q_1, q_2, q_3 and q_4 are selected as the **displacement of the springs K_1, K_3** and **velocity of the masses M_1, M_2** .

$$q_1 = x_1 \quad q_2 = \dot{x}_1 \quad q_3 = x_2 \quad q_4 = \dot{x}_2$$



State-Space Representation

- The following example illustrates how the number of state variables can be **less** than the number of energy-storing elements.

Example 4 Find the state-space model for the following LTI system.
Assume the applied force $f_a(t)$ as the input, and the displacement of spring K_2 is the output.

Mass $M_1 \rightarrow -K_1x_1 - K_2(x_1 - x_2) - B(\dot{x}_1 - \dot{x}_2) = M_1\ddot{x}_1$

Mass $M_2 \rightarrow f_a(t) - K_2(x_2 - x_1) - B(\dot{x}_2 - \dot{x}_1) - K_3x_2 = M_2\ddot{x}_2$

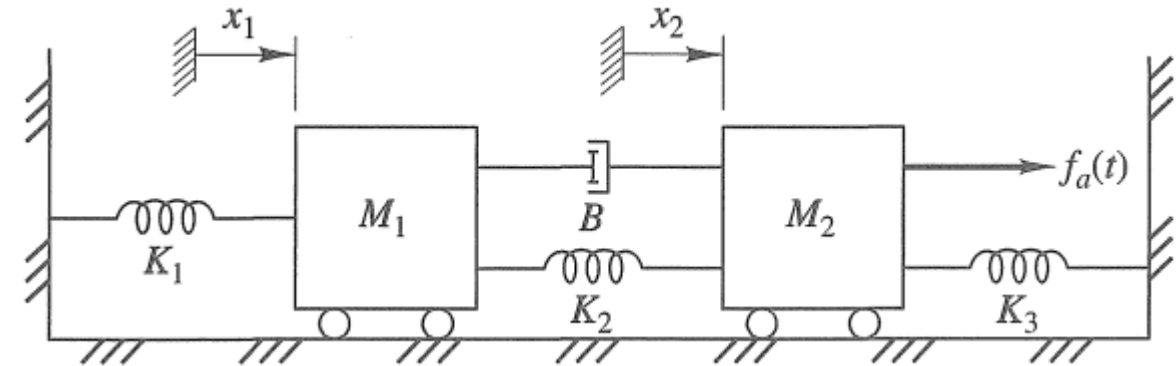
The **state-variable equations** are derived as:

$$q_1 = x_1 \rightarrow \dot{q}_1 = \dot{x}_1 \rightarrow \dot{q}_1 = q_2 \quad \text{Eqn. (1)}$$

$$\begin{aligned} q_2 = \dot{x}_1 \rightarrow \dot{q}_2 = \ddot{x}_1 \rightarrow \dot{q}_2 &= \frac{1}{M_1}(-K_1x_1 - K_2(x_1 - x_2) - B(\dot{x}_1 - \dot{x}_2)) \\ \rightarrow \dot{q}_2 &= \frac{1}{M_1}(-(K_1 + K_2)q_1 + K_2q_3 - B(q_2 - q_4)) \quad \text{Eqn. (2)} \end{aligned}$$

$$q_3 = x_2 \rightarrow \dot{q}_3 = \dot{x}_2 \rightarrow \dot{q}_3 = q_4 \quad \text{Eqn. (3)}$$

$$\begin{aligned} q_4 = \dot{x}_2 \rightarrow \dot{q}_4 = \ddot{x}_2 \rightarrow \dot{q}_4 &= \frac{1}{M_2}(f_a(t) - K_2(x_2 - x_1) - B(\dot{x}_2 - \dot{x}_1) - K_3x_2) \\ \rightarrow \dot{q}_4 &= \frac{1}{M_2}(f_a(t) - K_2(q_3 - q_1) - B(q_4 - q_2) - K_3q_3) \quad \text{Eqn. (4)} \end{aligned}$$



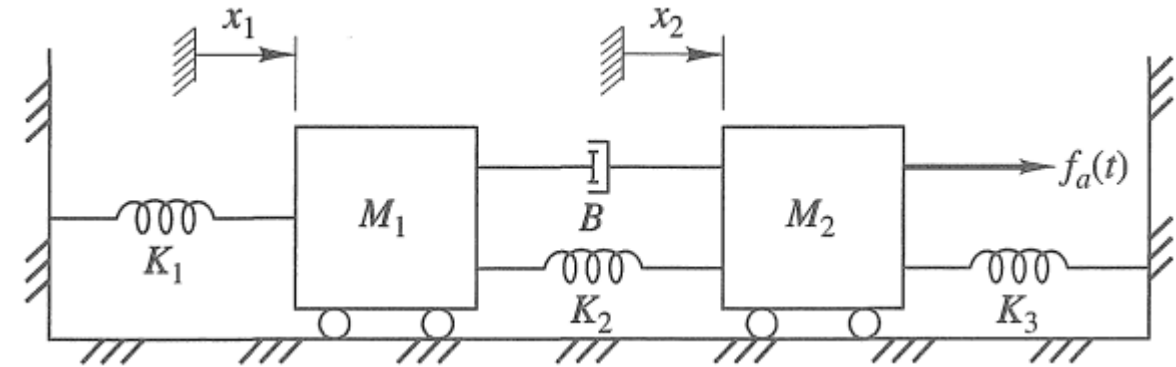
State-Space Representation

- The following example illustrates how the number of state variables can be **less** than the number of energy-storing elements.

Example 4 Find the state-space model for the following LTI system.
Assume the applied force $f_a(t)$ as the input, and the displacement of spring K_2 is the output.

The **state-variable equations** and **output equation** are obtained as:

$$\left\{ \begin{array}{l} \dot{q}_1 = q_2 \\ \dot{q}_2 = \frac{1}{M_1} (-(K_1 + K_2)q_1 - Bq_2 + K_2q_3 + Bq_4) \\ \dot{q}_3 = q_4 \\ \dot{q}_4 = \frac{1}{M_2} (K_2q_1 + Bq_2 - (K_2 + K_3)q_3 - Bq_4 + f_a(t)) \\ y = x_2 - x_1 \rightarrow y = q_3 - q_1 \end{array} \right.$$



We can represent the **state** and **output equations** in the standard matrix-vector form as below:

State Equation $\dot{\mathbf{q}}(t) = \mathbf{A} \mathbf{q}(t) + \mathbf{B} \mathbf{u}(t)$

$$\begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \\ \dot{q}_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -(K_1 + K_2)/M_1 & -B/M_1 & K_2/M_1 & B/M_1 \\ 0 & 0 & 0 & 1 \\ K_2/M_2 & B/M_2 & -(K_2 + K_3)/M_2 & -B/M_2 \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1/M_2 \end{bmatrix} f_a(t)$$

Output Equation $\mathbf{y}(t) = \mathbf{C} \mathbf{q}(t) + \mathbf{D} \mathbf{u}(t)$

$$y(t) = [-1 \quad 0 \quad 1 \quad 0] \begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{bmatrix} + [0]u(t)$$

State-Space Representation

- The following example, two springs and a damper are attached to a massless junction. The system contains three energy-storing elements and normally requires three state variables. However, when the damper is removed from the massless junction, the number of state variables is reduced.

Example 6 Find the state-space model for the following LTI system.
Assume the applied force $f_a(t)$ as the input, and the displacement of massless junction A is the output.

Draw the free-body diagram of the system and write the equation of motion of the system.

Mass $M \rightarrow$ $f_a(t) - K_1(x_1 - x_2) - B_1\dot{x}_1 = M\ddot{x}_1$

Junction A $\rightarrow$ $K_1(x_1 - x_2) = K_2x_2 + B_2\dot{x}_2$

The **state variables** q_1, q_2 and q_3 are selected as the **displacement of the springs K_1, K_2** and **velocity of the mass M** .

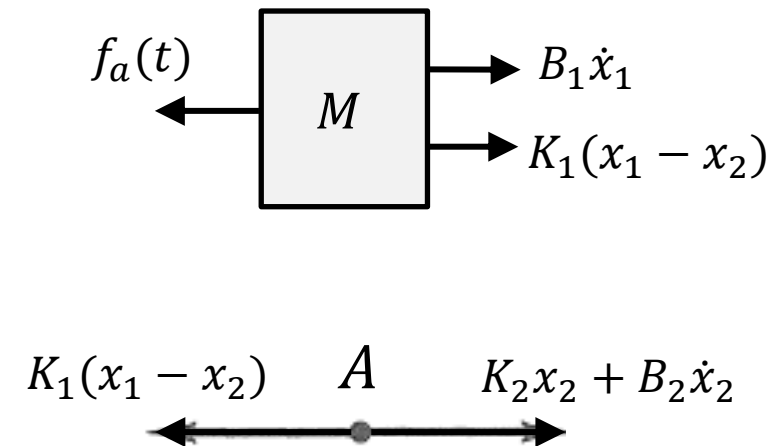
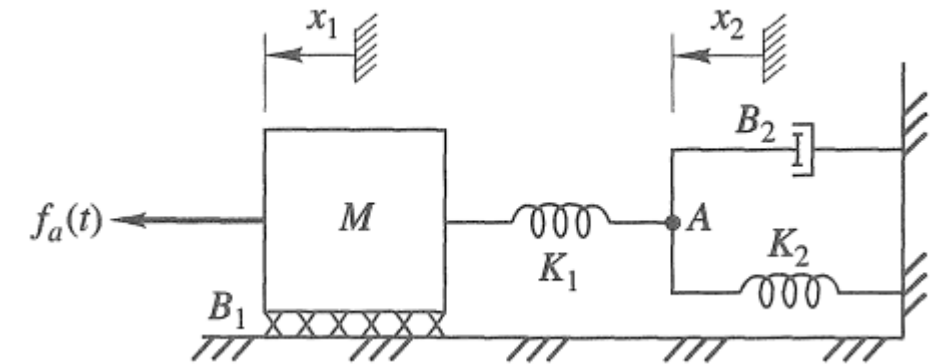
$$q_1 = x_1 - x_2 \rightarrow \dot{q}_1 = \dot{x}_1 - \dot{x}_2 \rightarrow \dot{q}_1 = q_3 - \frac{1}{B_2}(K_1q_1 - K_2q_2) \quad \text{Eqn. (1)}$$

$$q_2 = x_2 \rightarrow \dot{q}_2 = \dot{x}_2 \rightarrow \dot{q}_2 = \frac{1}{B_2}(K_1(x_1 - x_2) - K_2x_2)$$

$$\rightarrow \dot{q}_2 = \frac{1}{B_2}(K_1q_1 - K_2q_2) \quad \text{Eqn. (2)}$$

$$q_3 = \dot{x}_1 \rightarrow \dot{q}_3 = \ddot{x}_1 \rightarrow \dot{q}_3 = \frac{1}{M}(f_a(t) - K_1(x_1 - x_2) - B_1\dot{x}_1)$$

$$\rightarrow \dot{q}_3 = \frac{1}{M}(f_a(t) - K_1q_1 - B_1q_3) \quad \text{Eqn. (3)}$$



State-Space Representation

- The following example, two springs and a damper are attached to a massless junction. The system contains three energy-storing elements and normally requires three state variables. However, when the damper is removed from the massless junction, the number of state variables is reduced.

Example 6

Find the state-space model for the following LTI system.

Assume the applied force $f_a(t)$ as the input, and the displacement of massless junction A is the output.

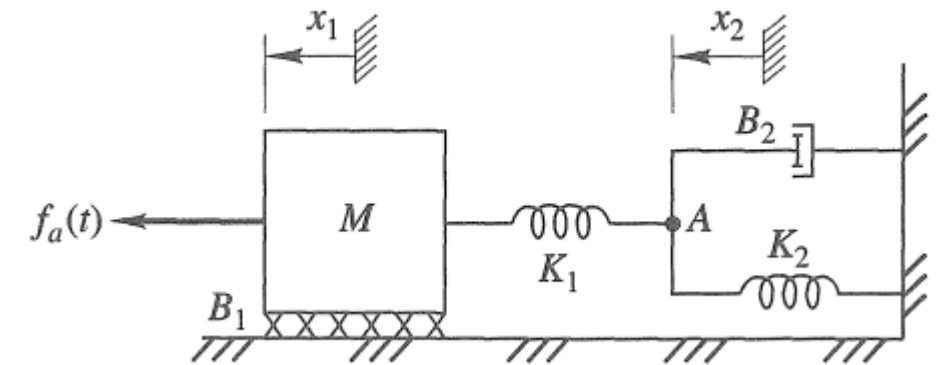
The state-variable equations and output equation are obtained as:

$$\begin{cases} \dot{q}_1 = -\frac{1}{B_2}(K_1 q_1 - K_2 q_2) + q_3 \\ \dot{q}_2 = \frac{1}{B_2}(K_1 q_1 - K_2 q_2) \\ \dot{q}_3 = \frac{1}{M}(f_a(t) - K_1 q_1 - B_1 q_3) \\ y = x_2 \rightarrow y = q_2 \end{cases}$$

We can represent the state and output equations in the standard matrix-vector form as below:

$$\dot{\mathbf{q}}(t) = \mathbf{A} \mathbf{q}(t) + \mathbf{B} \mathbf{u}(t) \quad \text{State Equation}$$

$$\begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{bmatrix} = \begin{bmatrix} -K_1/B_2 & K_2/B_2 & 1 \\ K_1/B_2 & -K_2/B_2 & 0 \\ -K_1/M & 0 & -B_1/M \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1/M \end{bmatrix} f_a(t)$$



Output Equation

$$\mathbf{y}(t) = \mathbf{C} \mathbf{q}(t) + \mathbf{D} \mathbf{u}(t)$$

$$y(t) = [0 \quad 1 \quad 0] \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} + [0] u(t)$$

State-Space Representation

- The following example, two springs and a damper are attached to a massless junction. The system contains three energy-storing elements and normally requires three state variables. However, when the damper is removed from the massless junction, the number of state variables is reduced.

Example 6

Find the state-space model for the following LTI system.

Assume the applied force $f_a(t)$ as the input, and the displacement of massless junction A is the output.

Removing the damper corresponds to setting $B_2 = 0$ in the equation of motion.

Mass $M \rightarrow f_a(t) - K_1(x_1 - x_2) - B_1\dot{x}_1 = M\ddot{x}_1$

Junction A $\rightarrow K_1(x_1 - x_2) = K_2x_2$

From the second equation $\rightarrow x_2 = \left(\frac{K_1}{K_1 + K_2} \right) x_1$

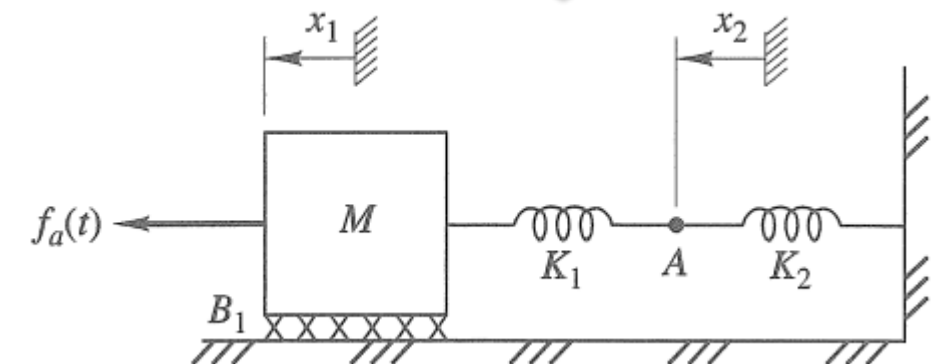
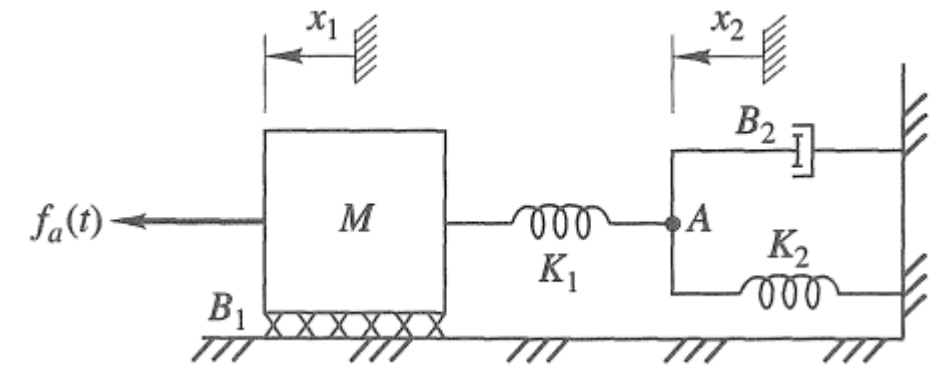
The displacements x_1 and x_2 are now **proportional** to each other, so they cannot both be state variables.

The two springs are now in **series** and can be replaced by a single **equivalent spring**,

$$K_{eq} = \frac{K_1 K_2}{K_1 + K_2}$$

The equation of motion can be rewritten as,

Mass $M \rightarrow f_a(t) - K_{eq}x_1 - B_1\dot{x}_1 = M\ddot{x}_1$



State-Space Representation

- The following example, two springs and a damper are attached to a massless junction. The system contains three energy-storing elements and normally requires three state variables. However, when the damper is removed from the massless junction, the number of state variables is reduced.

Example 6

Find the state-space model for the following LTI system.

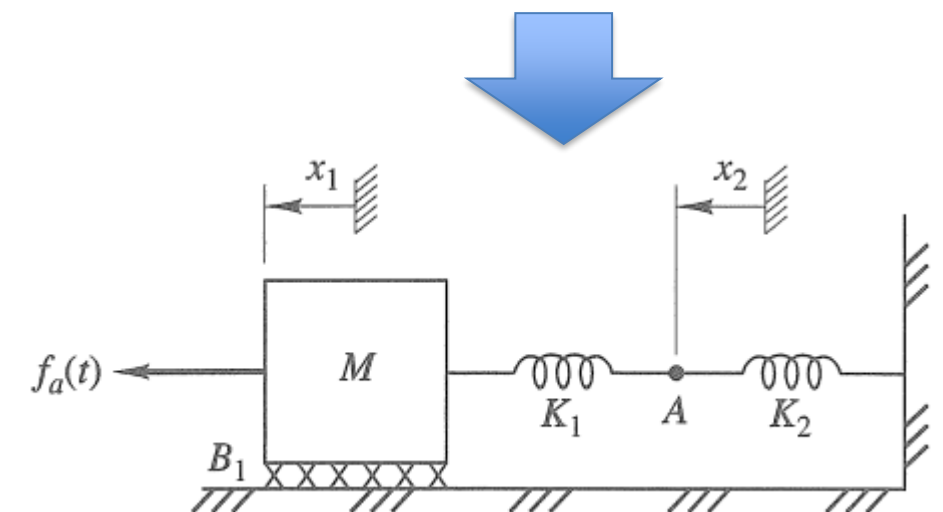
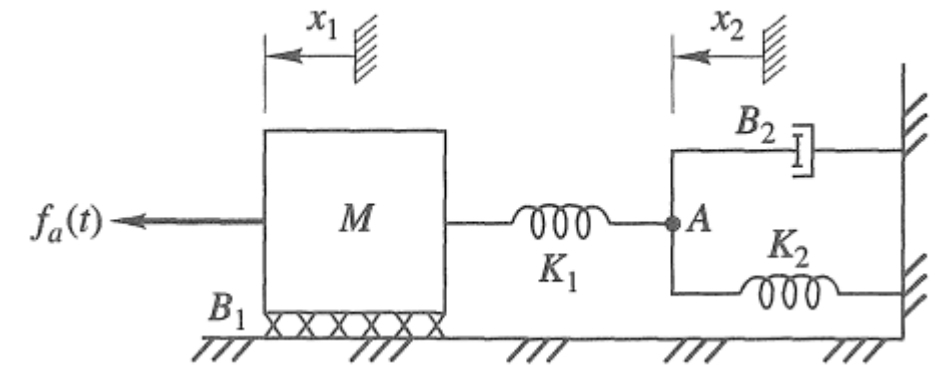
Assume the applied force $f_a(t)$ as the input, and the displacement of massless junction A is the output.

Mass $M \rightarrow f_a(t) - K_{eq}x_1 - B_1\dot{x}_1 = M\ddot{x}_1$

The state variables q_1 and q_2 are selected as the displacement of the equivalent spring K_{eq} and velocity of the mass M .

$$q_1 = x_1 \rightarrow \dot{q}_1 = \dot{x}_1 \rightarrow \dot{q}_1 = q_2 \quad \text{Eqn. (1)}$$

$$\begin{aligned} q_2 = \dot{x}_1 \rightarrow \dot{q}_2 = \ddot{x}_1 \rightarrow \dot{q}_2 &= \frac{1}{M}(f_a(t) - K_{eq}x_1 - B_1\dot{x}_1) \\ \rightarrow \dot{q}_2 &= \frac{1}{M}(f_a(t) - K_{eq}q_1 - B_1q_2) \quad \text{Eqn. (2)} \end{aligned}$$



State-Space Representation

- The following example, two springs and a damper are attached to a massless junction. The system contains three energy-storing elements and normally requires three state variables. However, when the damper is removed from the massless junction, the number of state variables is reduced.

Example 6 Find the state-space model for the following LTI system.
Assume the applied force $f_a(t)$ as the input, and the displacement of massless junction A is the output.

The state-variable equations and output equation are obtained as:

$$\begin{cases} \dot{q}_1 = q_2 \\ \dot{q}_2 = \frac{1}{M} \left(f_a(t) - \frac{K_1 K_2}{K_1 + K_2} q_1 - B_1 q_2 \right) \\ y = x_2 \rightarrow y = \left(\frac{K_1}{K_1 + K_2} \right) x_1 \rightarrow y = \left(\frac{K_1}{K_1 + K_2} \right) q_1 \end{cases}$$

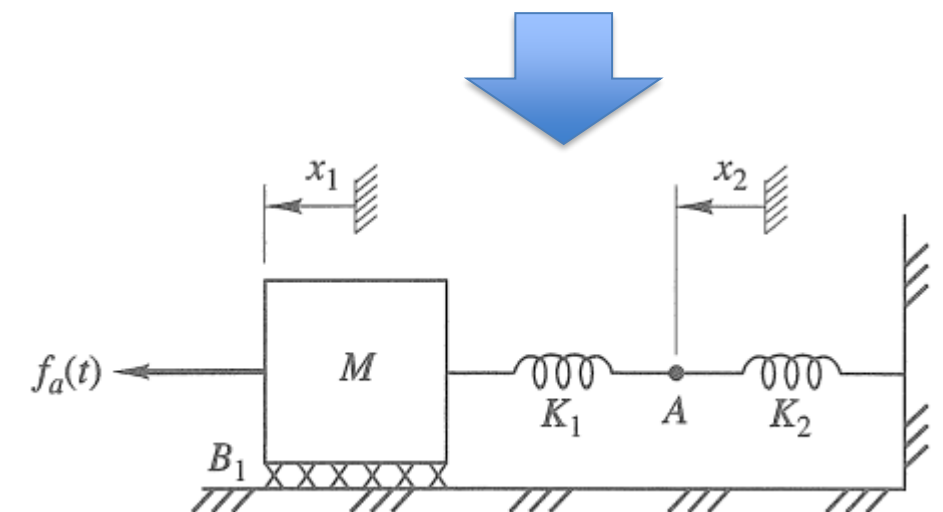
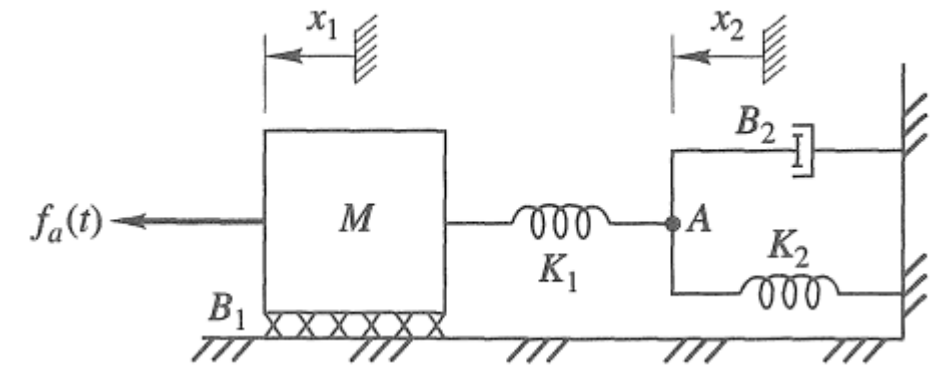
We can represent the state and output equations in the standard matrix-vector form as below:

State Equation

$$\begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \frac{-K_1 K_2}{M(K_1 + K_2)} & -\frac{B_1}{M} \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{M} \end{bmatrix} f_a(t)$$

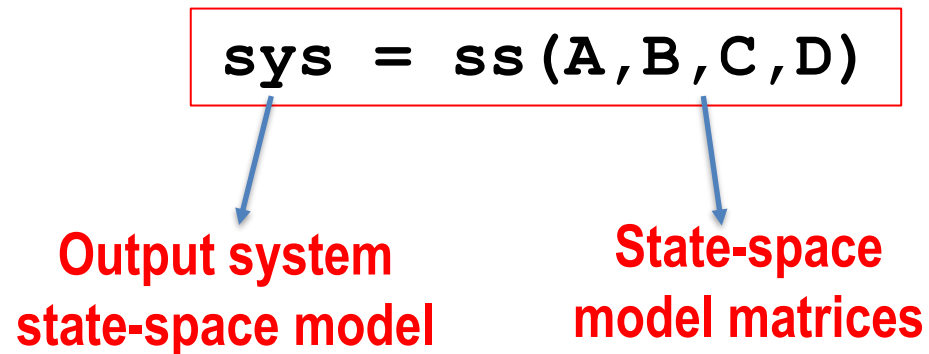
Output Equation

$$y(t) = \begin{bmatrix} \frac{K_1}{K_1 + K_2} & 0 \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} + [0] f_a(t)$$



State-Space Representation with MATLAB

- We can create the **State-Space Representation** of a continuous-time system with **MATLAB** by using the following command:



- For example, create the given state-space model in MATLAB.

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} u(t)$$

```
A = [0 1;-2 -3];  
B = [0;1];  
C = [1 0];  
D = 0;  
sys = ss(A,B,C,D)  
sys =
```

```
A =  
      x1  x2  
x1      0   1  
x2     -2  -3
```

```
B =  
      u1  
x1      0  
x2      1
```

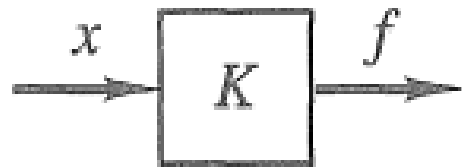
```
C =  
      x1  x2  
y1      1   0
```

```
D =  
      u1  
y1      0
```

Continuous-time state-space model.

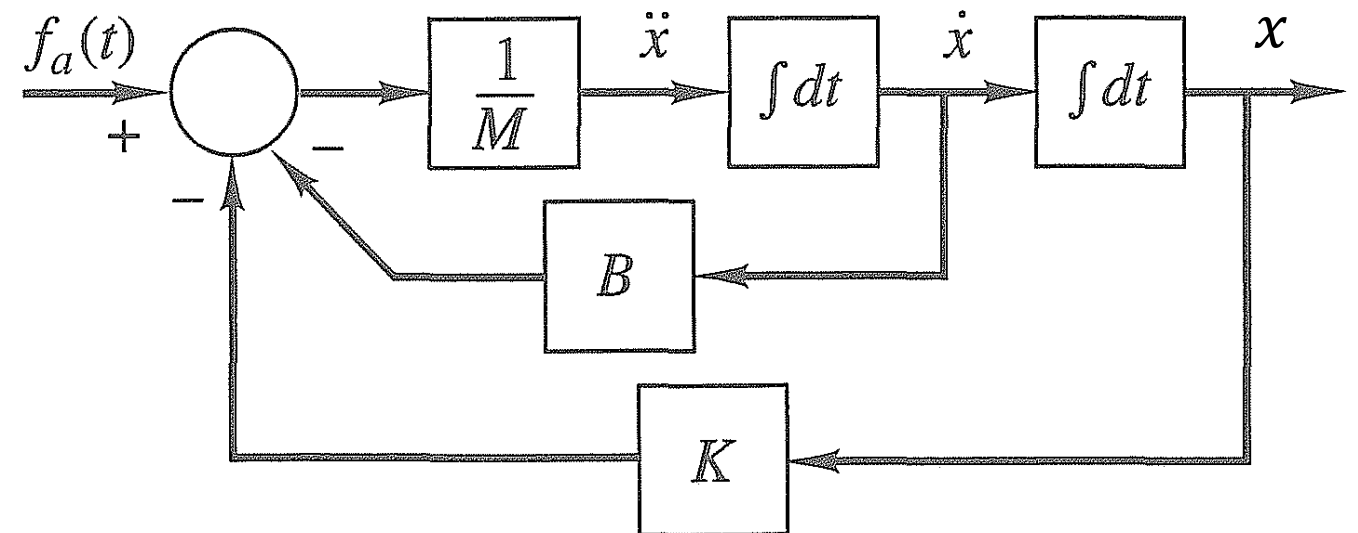
Block Diagram Models

- A **block diagram model** is an **interconnection of blocks** representing basic mathematical operations in such a way that the overall diagram is **equivalent** to the system's mathematical model.
- In a block diagram model,
 - The **arrows** interconnecting the blocks represent the **variables** describing the system behavior and **flow** of the signals.
 - For example, inputs, outputs, state variables,
 - The **blocks** represents **operations** or **functions** that use one or more of these variables to calculate other variables.
 - For example, in equation $f = Kx$ the force can be calculated as,



□ Basic Elements of Block Diagram:

- Addition/Subtraction Block
- Gain Block
- Integrator Block
- Constant Block

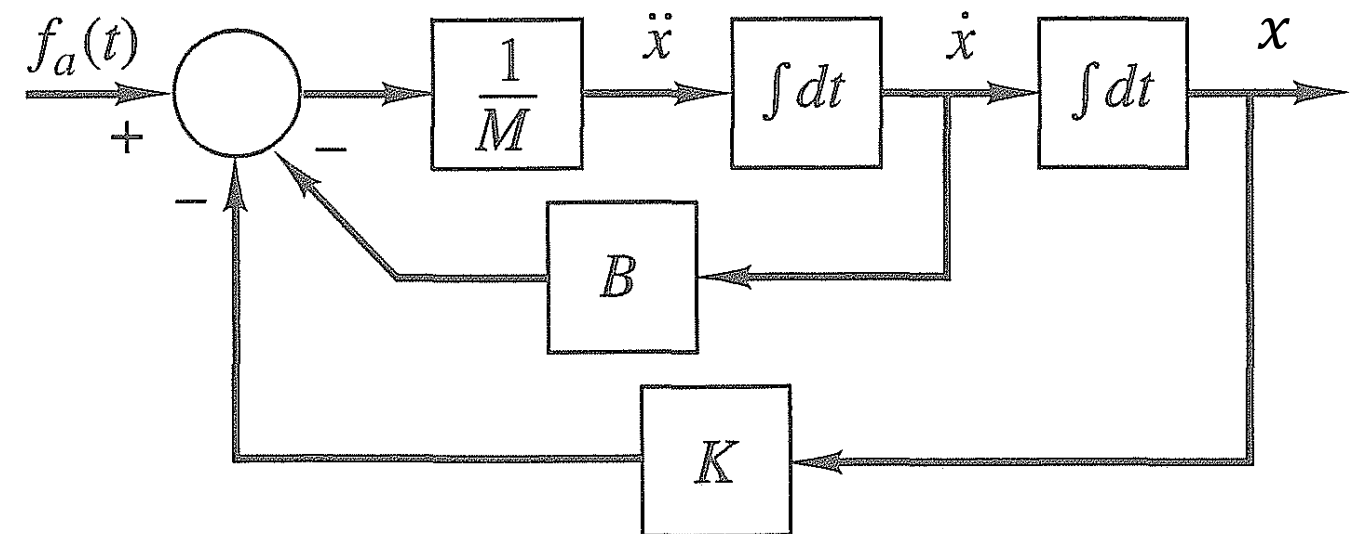


Block Diagram Models from Input-Output Equations

- We will show that how to combine the block diagram elements, such as gains, summer, integrator, and constant value blocks, to represent and solve an input-output differential equation model of a system.

□ Find the Block Diagrams Model for Input-Output Equations

- Identify the output variables.
- Solve the given equation for the highest derivative of each output variables
- Create a sequence of output variables using integrator block to relate $\ddot{x}$, $\dot{x}$, and x to one another.
- Complete the diagram with addition/subtraction and gain blocks according to the model equation



Block Diagram Models from Input-Output Equations

Example 7

Construct a block diagram model for the following system, whose input-output model was given as below:

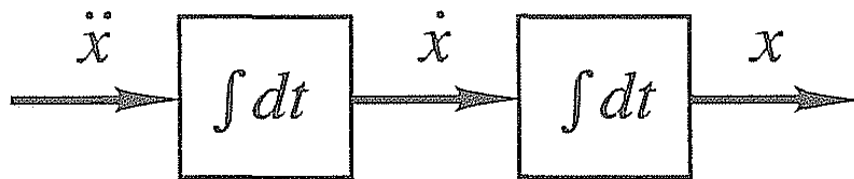
$$M\ddot{x}(t) + B\dot{x}(t) + Kx(t) = f_a(t)$$

1) Identify the output variable $\rightarrow x(t)$

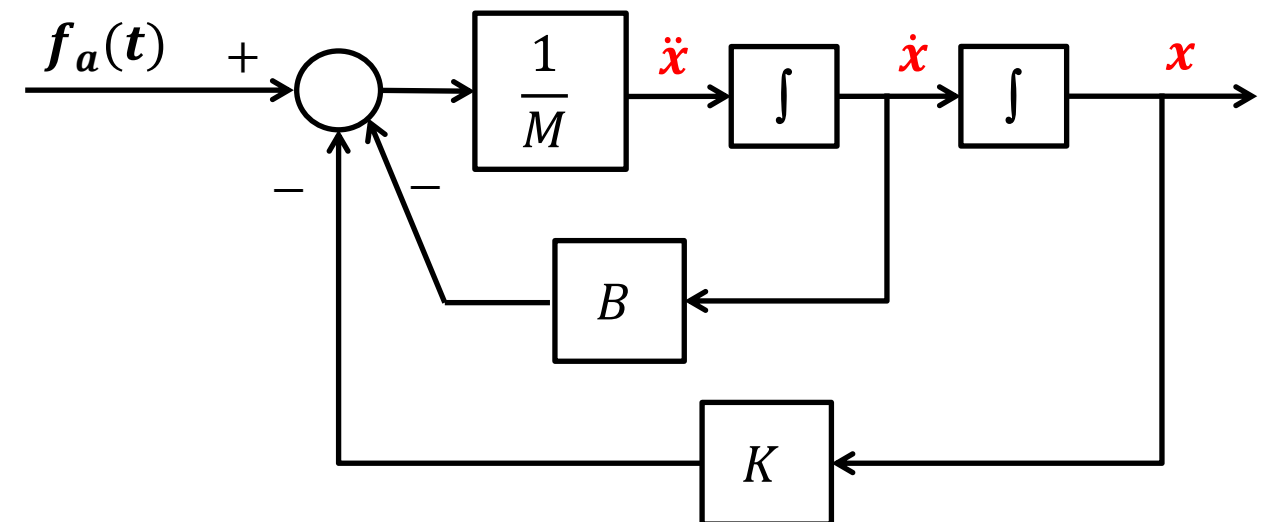
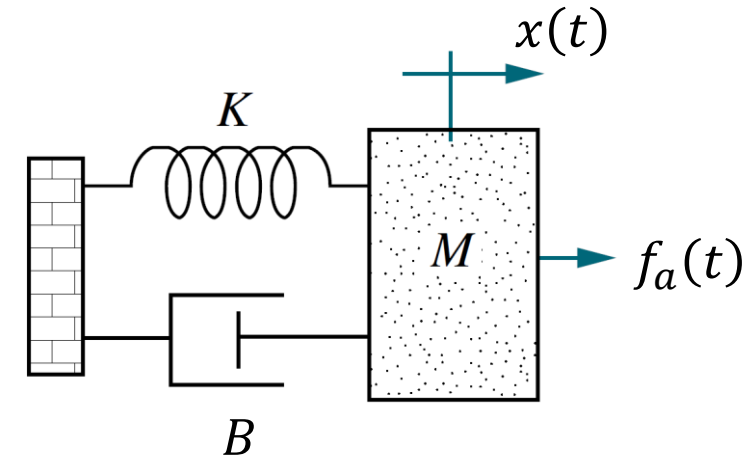
2) Solve for highest derivative of output variable

$$\ddot{x}(t) = \frac{1}{M}(f_a(t) - B\dot{x}(t) - Kx(t))$$

3) Create a sequence of output variables using integrator block to relate $\ddot{x}$, $\dot{x}$, and x to one another.



4) Complete the diagram with addition/subtraction and gain blocks according to the model equation

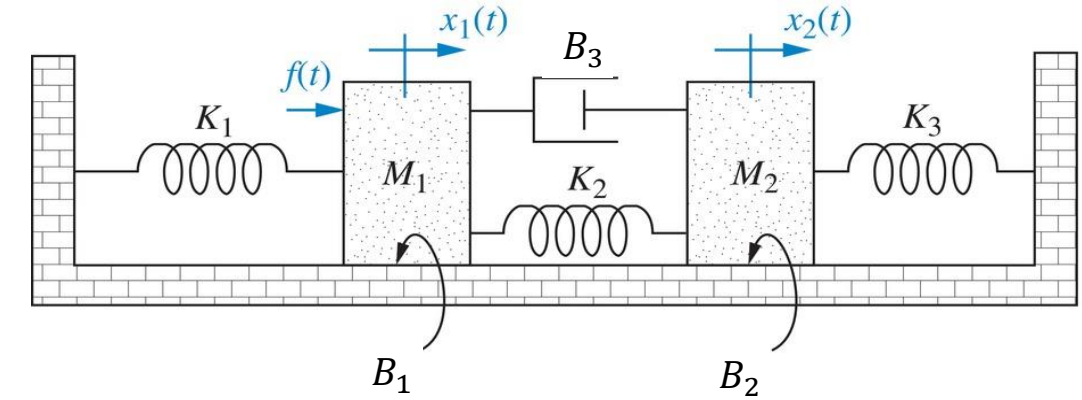


Block Diagram Models from Input-Output Equations

Example 8

Construct a block diagram model for the following system, whose input-output model was given as below:

$$\begin{cases} M_1 \ddot{x}_1 + (B_1 + B_3) \dot{x}_1 + (K_1 + K_2) x_1 - B_3 \dot{x}_2 - K_2 x_2 = f(t) \\ M_2 \ddot{x}_2 + (B_2 + B_3) \dot{x}_2 + (K_2 + K_3) x_2 - B_3 \dot{x}_1 - K_2 x_1 = 0 \end{cases}$$



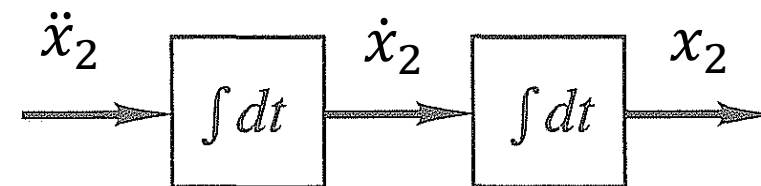
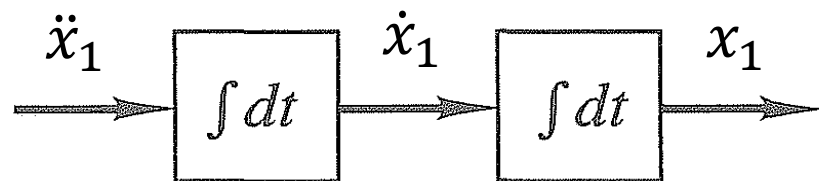
1) Identify the output variables $\rightarrow x_1(t)$ and $x_2(t)$

2) Solve for highest derivative of output variables

$$\ddot{x}_1 = \frac{1}{M_1} (f(t) - (B_1 + B_3) \dot{x}_1 - (K_1 + K_2) x_1 + B_3 \dot{x}_2 + K_2 x_2)$$

$$\ddot{x}_2 = \frac{1}{M_2} (-(B_2 + B_3) \dot{x}_2 - (K_2 + K_3) x_2 + B_3 \dot{x}_1 + K_2 x_1)$$

3) Create a sequence of output variables using integrator block to relate $\ddot{x}$, $\dot{x}$, and x to one another.

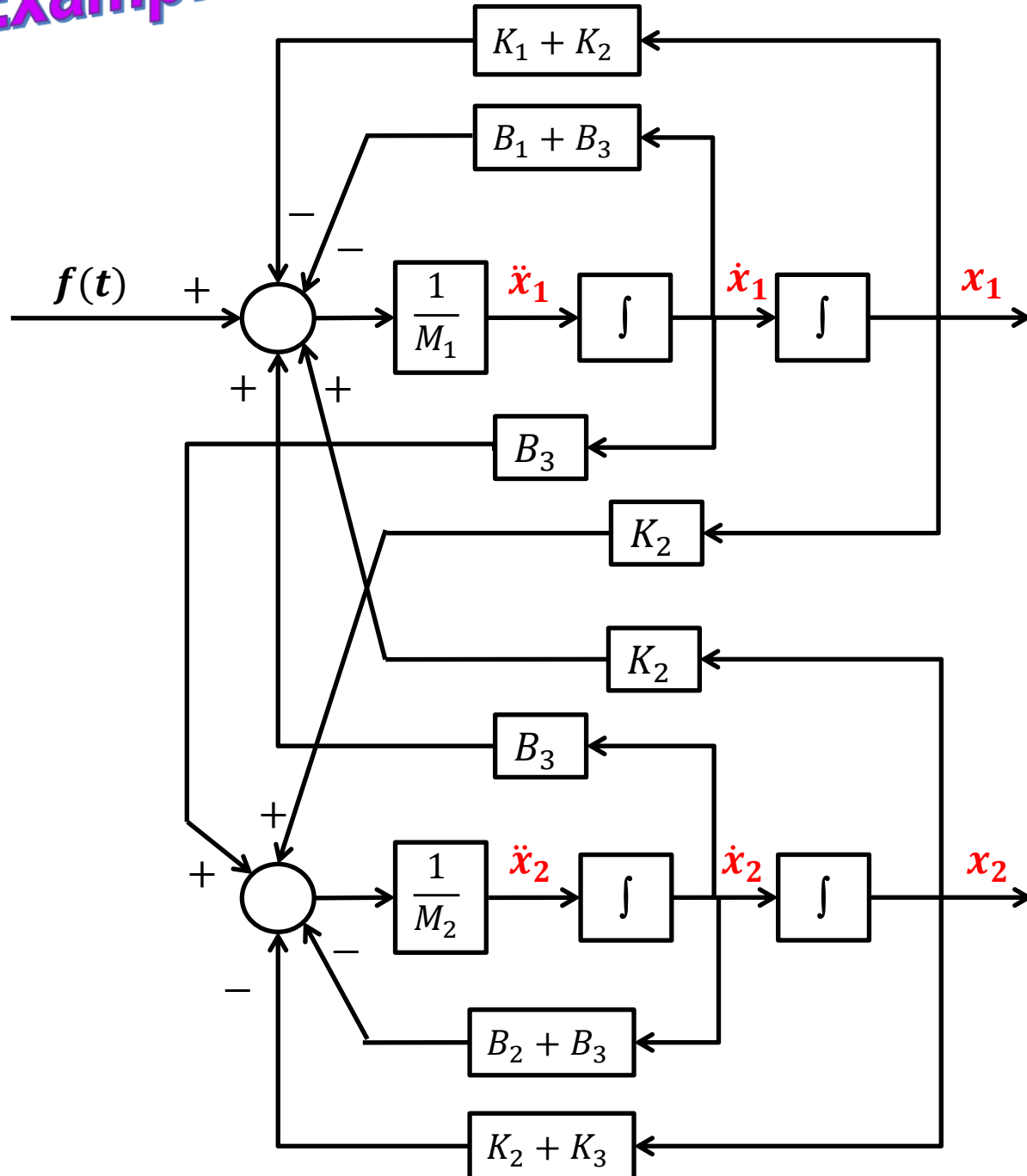


Block Diagram Models from Input-Output Equations

Example 8

Construct a block diagram model for the following system, whose input-output model was given as below:

4) Complete the diagram with addition/subtraction and gain blocks.



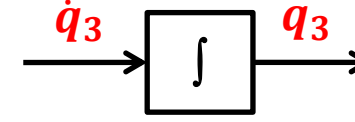
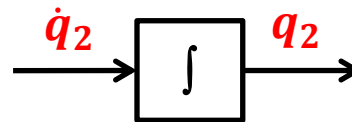
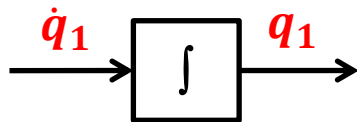
$$\ddot{x}_1 = \frac{1}{M_1} (f(t) - (B_1 + B_3)\dot{x}_1 - (K_1 + K_2)x_1 + B_3\dot{x}_2 + K_2x_2)$$

$$\ddot{x}_2 = \frac{1}{M_2} (-(B_2 + B_3)\dot{x}_2 - (K_2 + K_3)x_2 + B_3\dot{x}_1 + K_2x_1)$$

Block Diagram Models from State-Space Equations

□ Find the Block Diagram Model for State-Variable Models

- To construct a block diagram for a state-variable model:
 - 1) Draw an **integrator** for each **state variable**.
 - The inputs to these integrators are the **first derivatives** of the corresponding state variables.

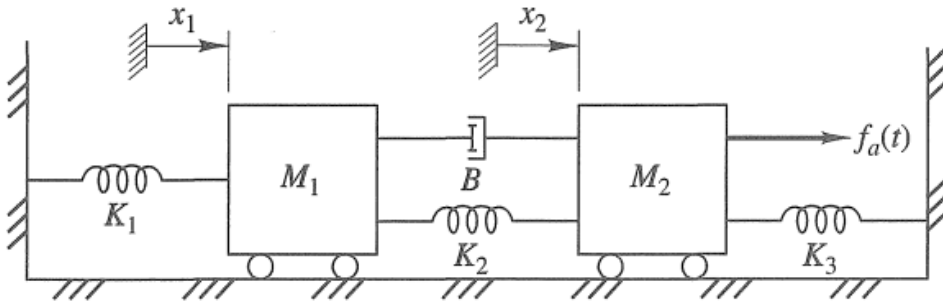


- 2) Assemble terms needed to create each **state variable derivative**, using the **state-variable equations**.
 - These will be functions of state variables and inputs.
- 3) Assemble the blocks needed for **output variable**, using the **output equation**.

Block Diagram Models from State-Space Equations

Example 9

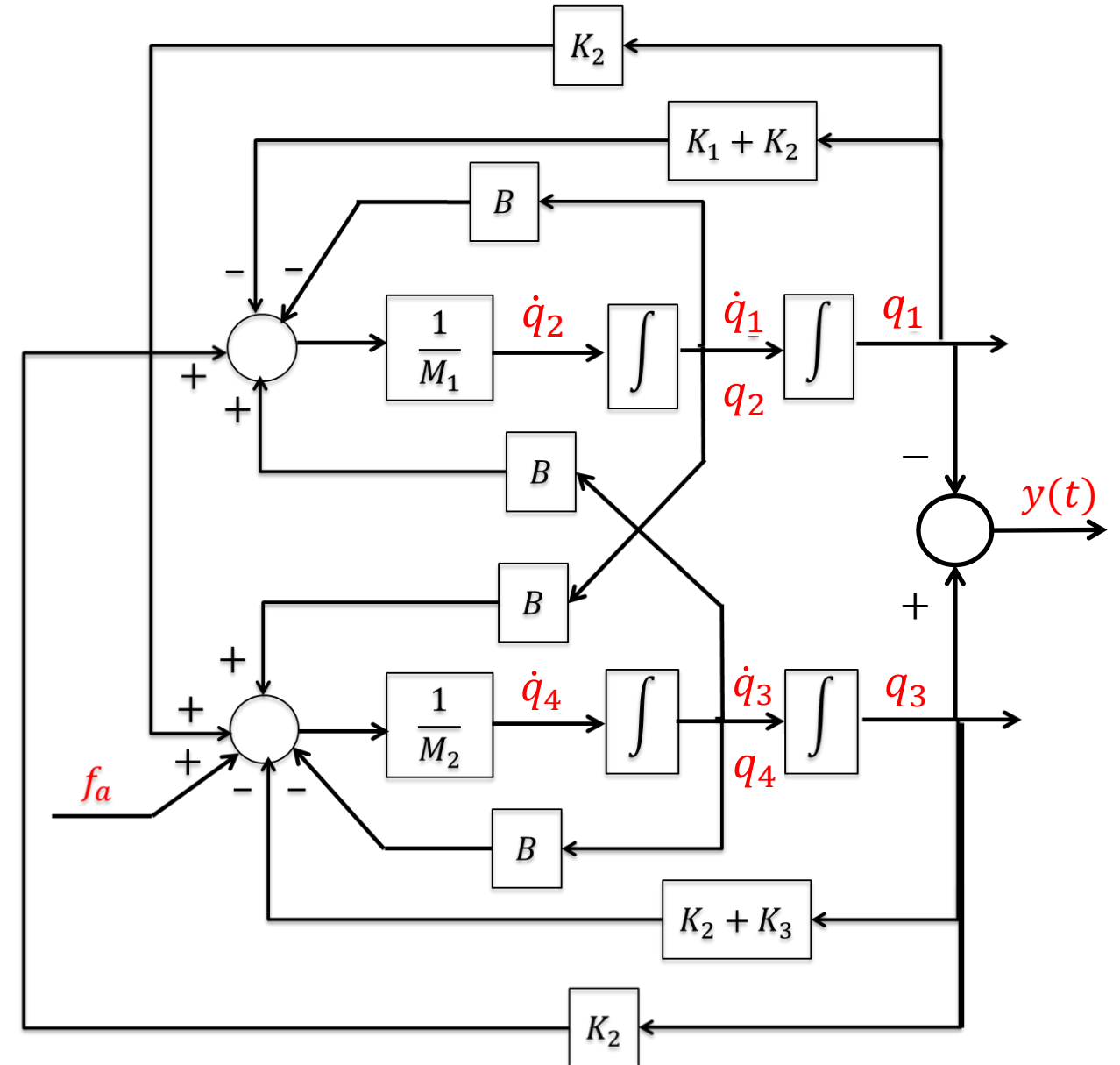
Recall the mechanical system from Example 4. Given the state-variable model draw the block diagram model of the system.



$$\begin{aligned} q_1 &= x_1 \\ q_2 &= \dot{x}_1 \\ q_3 &= x_2 \\ q_4 &= \dot{x}_2 \end{aligned}$$

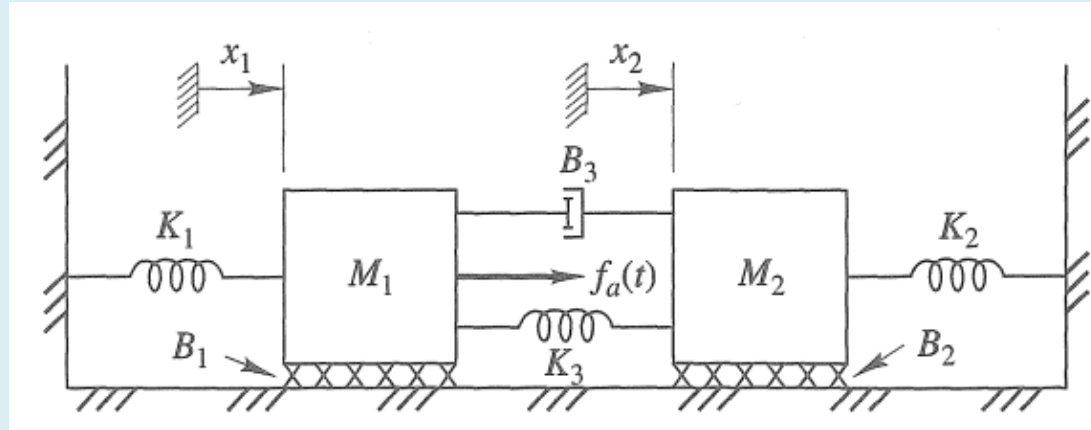
$$\begin{cases} \dot{q}_1(t) = q_2(t) \\ \dot{q}_2(t) = \frac{1}{M_1} (-(K_1 + K_2)q_1 - Bq_2 + K_2q_3 + Bq_4) \\ \dot{q}_3(t) = q_4(t) \\ \dot{q}_4(t) = \frac{1}{M_2} (f_a(t) - (K_2 + K_3)q_3 - Bq_4 + K_2q_1 + Bq_2) \\ y(t) = x_2 - x_1 \rightarrow y = q_3 - q_1 \end{cases}$$

- 1) Draw an integrator for each state variable
- 2) Assemble terms needed to create each state variable derivative
- 3) Assemble the blocks needed for any output equations



Quick Review

Draw a block diagram model to represent the state-variable of the system shown below from the given state-variable equations.



The state-variables are defined as:

$$q_1 = x_1 \quad q_2 = \dot{x}_1 \quad q_3 = x_2 \quad q_4 = \dot{x}_2$$

The state-variable equations and output equation are obtained as:

$$\begin{cases} \dot{q}_1 = q_2 \\ \dot{q}_2 = \frac{1}{M_1} (f_a(t) - (K_1 + K_3)q_1 - (B_1 + B_3)q_2 + K_3q_3 + B_3q_4) \\ \dot{q}_3 = q_4 \\ \dot{q}_4 = \frac{1}{M_2} (K_3q_1 + B_3q_2 - (K_2 + K_3)q_3 - (B_2 + B_3)q_4) \\ y = x_2 \rightarrow y = q_3 \end{cases}$$

THANK YOU